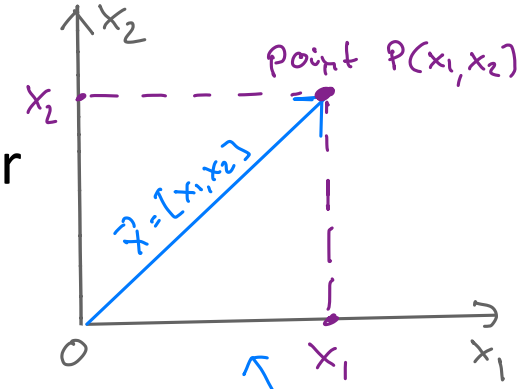
III. Linear Programming

Notation: In the following $\mathbf{x}$ or also $\vec{x}$ will denote a point or vector in $\mathbb{R}^n$,

when handwritten we use the arrow →

$$\mathbf{x} = (x_1, x_2, x_3, \ldots, x_n) \quad \text{--- a point}$$

$$\vec{x} = \mathbf{x} = [x_1, x_2, x_3, \ldots, x_n]^T \quad \text{--- a vector}$$



where $x_1, x_2, x_3, \ldots, x_n$ are the coordinates (components) of $\mathbf{x}$.

a point

e.g. $\mathbf{x} = (\,1.2,\ 3.43,\ -1,\ 0\,)$ or $\mathbf{0} = (\,0,\ 0,\ 0,\ 0\,)$

$$\vec{x} = \begin{bmatrix} 1.2 \\ 3.43 \\ -1 \\ 0 \end{bmatrix}$$ a vector

We identify the point (x_1, x_2) with its position vector $\vec{x} = [x_1, x_2]$ (or $\vec{x} = \langle x_1, x_2 \rangle$)

For two vectors $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$,

$\mathbf{x} \le \mathbf{y}$ means $x_i \le y_i$ for $i = 1, 2, 3$.

$\vec{x} \le \vec{y}$ means: $x_1 \le y_1$ and $x_2 \le y_2$ and $x_3 \le y_3$

(In a similar way, we define $\mathbf{x} < \mathbf{y}$, $\mathbf{x} \geq \mathbf{y}$, $\mathbf{x} > \mathbf{y}$)

$$\vec{x} = \mathbf{x} = \begin{bmatrix} 2 \\ -0.4 \\ -2.4 \\ 3.6 \end{bmatrix} \quad \begin{matrix} 2 < 4 \\ -0.4 < -0.2 \\ -2.4 < 0.4 \\ 3.6 < 7.8 \end{matrix} \quad \vec{y} = \mathbf{y} = \begin{bmatrix} 4 \\ -0.2 \\ 0.4 \\ 7.8 \end{bmatrix}.$$

Then $\vec{x} < \vec{y}$ and $\vec{x} \leq \vec{y}$ are <u>both</u> true

$$\vec{a} = \mathbf{a} = \begin{bmatrix} -2 \\ 4 \\ 0 \end{bmatrix} = [-2, 4, 0]^T \qquad \vec{b} = \mathbf{b} = \begin{bmatrix} -1 \\ 3 \\ 1 \end{bmatrix} = [-1, 3, 1]^T$$

$-2 < -1$

$4 > 3$

Thus,

So $\vec{a} \leq \vec{b}$ is <u>false</u>

$\vec{a} \geq \vec{b}$ is <u>false</u> as well !!

We can <u>not</u> compare $\vec{a}$ and $\vec{b}$

Vectors will be column vectors, so their transposed will be row vectors.

$$\vec{a} = \begin{bmatrix} -2 \\ 4 \\ 0 \end{bmatrix}$$

Column vector

or

$$\vec{a}^T = \begin{bmatrix} -2 & 4 & 0 \end{bmatrix}$$

$$\vec{a}^T = [-2, 4, 0]$$

row vector

Let's look at some practical examples introducing linear programming.

Example LP1: A company makes 4 different products: X_1, X_2, X_3, X_4.

There are 3 inputs: --- labor (in person-weeks)

--- raw material A (in kg) -- to make the products

--- material B (in boxes) – to make the products

The following table shows the inputs for one unit of each product, and the selling price for one unit of each product.

		Product				Input availabilities
		X_1	X_2	X_3	X_4	
Inputs per one Item produced	*Person-weeks*	1	2	1	2	20 (1)
	kg of material A	6	5	3	2	100 (2)
	Boxes of material B	3	4	9	12	75 (3)
sales price per one product item (in US$)		6	4	7	5	
Production levels (units)		x_1	x_2	x_3	x_4	

X_i = amount of product X_i produced

The <u>total revenue</u> for any production decision is therefore $c^T = [6, 4, 7, 5]$

$$f(x_1, x_2, x_3, x_4) = 6x_1 + 4x_2 + 7x_3 + 5x_4$$

However, the limited availability of inputs impose some constraints:

Person-weeks: $x_1 + 2x_2 + x_3 + 2x_4 \leq 20$ (1)

kg of material A: $6x_1 + 5x_2 + 3x_3 + 2x_4 \leq 100$ (2)

Boxes of material B: $3x_1 + 4x_2 + 9x_3 + 12x_4 \leq 75$ (3)

b

These coefficients make up the matrix A

Also, because one can only produce a positive number of products,

$$x_i \geq 0, \qquad i = 1, 2, 3, 4. \qquad (4)$$

The problem to be solved: Decide how many units of each product to produce in order to <u>maximize</u> the revenue $f(x_1, x_2, x_3, x_4)$

In matrix form:

$$\boxed{f(\vec{x}) = f(x_1, x_2, x_3, x_4) = \vec{c}^T \vec{x}, \qquad A\vec{x} \leq \vec{b} \qquad \vec{x} \geq 0}$$

where

$$\vec{c}^T = [6, 4, 7, 5] \qquad \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 6 & 5 & 3 & 2 \\ 3 & 4 & 9 & 12 \end{bmatrix} \qquad \vec{b} = \begin{bmatrix} 20 \\ 100 \\ 75 \end{bmatrix}$$

Remarks: 1. Any point $\mathbf{x} = (x_1, x_2, x_3, x_4)$ (or vector $\mathbf{x} = [x_1, x_2, x_3, x_4]^T$) in 4-dimensional space $\mathbb{R}^4$ that satisfies the constraints (1)—(4) is called a *feasible point* of *feasible solution.* All feasible points together form a subset Ω of $\mathbb{R}^4$ which is called the *feasible region* or *feasible domain*. The function $f(x_1, x_2, x_3, x_4)$ is called the *objective function.* Our task is

$$\text{maximize} \quad f(\mathbf{x})$$

$$\text{subject to} \quad \mathbf{x} \in \Omega$$

where $\mathbf{x} = (x_1, x_2, x_3, x_4)$.

2. The function f is called a *linear function* and the constraints (1)—(4) are called *linear (inequality) constraints*, as they only involve linear combinations of the variables x_1, x_2, x_3, x_4. Thus, the above problem is called a *linear programming problem*.

3. Any linear programming problem can be written in vector/matrix form. The above problem is

LP

$$\text{Maximize} \quad f(\mathbf{x}) = \mathbf{c}^T\mathbf{x}$$

$$\text{Subject to} \quad A\mathbf{x} \le \mathbf{b}$$

$$\mathbf{x} \ge \mathbf{0}$$

where $\mathbf{c}^T = [\,6, 4, 7, 5\,]$,

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 6 & 5 & 3 & 2 \\ 3 & 4 & 9 & 12 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 20 \\ 100 \\ 75 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

these are C(1), C(2), C(3)

Note:

$$A\vec{x} = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 6 & 5 & 3 & 2 \\ 3 & 4 & 9 & 12 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \underbrace{\begin{bmatrix} x_1 + 2x_2 + x_3 + 2x_4 \\ 6x_1 + 5x_2 + 3x_3 + 2x_4 \\ 3x_1 + 4x_2 + 9x_3 + 12x_4 \end{bmatrix}}_{A\vec{x}} \le \underbrace{\begin{bmatrix} 20 \\ 100 \\ 75 \end{bmatrix}}_{\vec{b}}$$

Example LP2: A company makes 2 different products, X_1, X_2, using 3 machines: M_1, M_2, M_3. The production time for each machine is as follows:

Machine	Production time (hours/unit)		Available time (hours) per machine
	X_1	X_2	
M_1	1	1	8
M_2	1	3	18
M_3	2	1	14
(*) – total Machine hours required per unit	4	5	
Production (units)	x_1	x_2	

x_i = number of units of X_i produced

Objective: Maximize the combined utilization time of all machines.

The objective function is

(*) says: it takes 4 hours utilization to produce one unit of X_1
it takes 5 hours utilization to produce one unit of X_2

$$f(x_1, x_2) = 4x_1 + 5x_2$$

with constraints [max. utilization time for each machine]

$$M_1: \quad x_1 + x_2 \le 8$$

$$M_2: \quad x_1 + 3x_2 \le 18$$

$$M_3: \quad 2x_1 + x_2 \le 14$$

$$x_1, x_2 \ge 0$$

In matrix/vector notation:

$$\text{Maximize} \quad f(\mathbf{x}) = \mathbf{c}^T \mathbf{x}$$

$$\text{Subject to} \quad A\mathbf{x} \le \mathbf{b}$$

$$\mathbf{x} \ge \mathbf{0}$$

where $\mathbf{c}^T = [4 \quad 5]$, $A = \begin{bmatrix} 1 & 1 \\ 1 & 3 \\ 2 & 1 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 8 \\ 18 \\ 4 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

III.1 Two-Dimensional Linear Programming (LP) Problems

We first consider LP-problems in $\mathbb{R}^2$, i.e. of two variables x_1, x_2, because we can understand the solutions using graphs.

Example LP3: Consider the problem:

$$\text{Maximize} \quad f(\mathbf{x}) = \mathbf{c}^T\mathbf{x} \longrightarrow \begin{bmatrix} 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1 + 5x_2$$

$$\text{Subject to} \quad A\mathbf{x} \le \mathbf{b} \longrightarrow \begin{bmatrix} 5 & 6 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5x_1 + 6x_2 \\ 3x_1 + 2x_2 \end{bmatrix} \le \begin{bmatrix} 30 \\ 12 \end{bmatrix}$$

$$\mathbf{x} \ge \mathbf{0} \longrightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \ge \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

where $\mathbf{c}^T = [\,1, 5\,]$, $\quad A = \begin{bmatrix} 5 & 6 \\ 3 & 2 \end{bmatrix}$, $\quad \mathbf{b} = \begin{bmatrix} 30 \\ 12 \end{bmatrix}$, $\quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

That is, the objective function is $f(x_1, x_2) = x_1 + 5x_2$

with constraints

① $5x_1 + 6x_2 \le 30$

② $3x_1 + 2x_2 \le 12$

$x_1 \ge 0 \quad x_2 \ge 0$

(x_1, x_2) is in the 1st quadrant

(1) The graph of a linear equation in 2 dimensions

(*) $\boxed{a x_1 + b x_2 = K}$ is a <u>straight line</u>

If $b = 0$, then

$$b x_2 = K - a x_1$$

$$\boxed{x_2 = \frac{K}{b} - \frac{a}{b} x_1}$$

x_2-intercepts slope $-\frac{a}{b}$

How to sketch this line:

(1) $x_1 = 0$ in (*) $\Rightarrow b x_2 = K \Rightarrow \boxed{x_2 = \frac{K}{b}}$ x_2-intercept

(2) $x_2 = 0$ in (*) $\Rightarrow a x_1 = K \Rightarrow \boxed{x_1 = \frac{K}{a}}$ x_1-intercept

(if $a \neq 0$)

Line: $5x_1 + 6x_2 = 30$:

x_1 - intercept: $(x_2 = 0) \Rightarrow 5x_1 = 30 \Rightarrow \underline{\underline{x_1 = 6}}$

x_2 - intercept: $(x_1 = 0) \Rightarrow 6x_2 = 30 \Rightarrow \underline{\underline{x_2 = 5}}$

Line: $3x_1 + 2x_2 = 12$

x_1 - intercept: $(x_2 = 0) \Rightarrow 3x_1 = 12 \Rightarrow \underline{\underline{x_1 = 4}}$

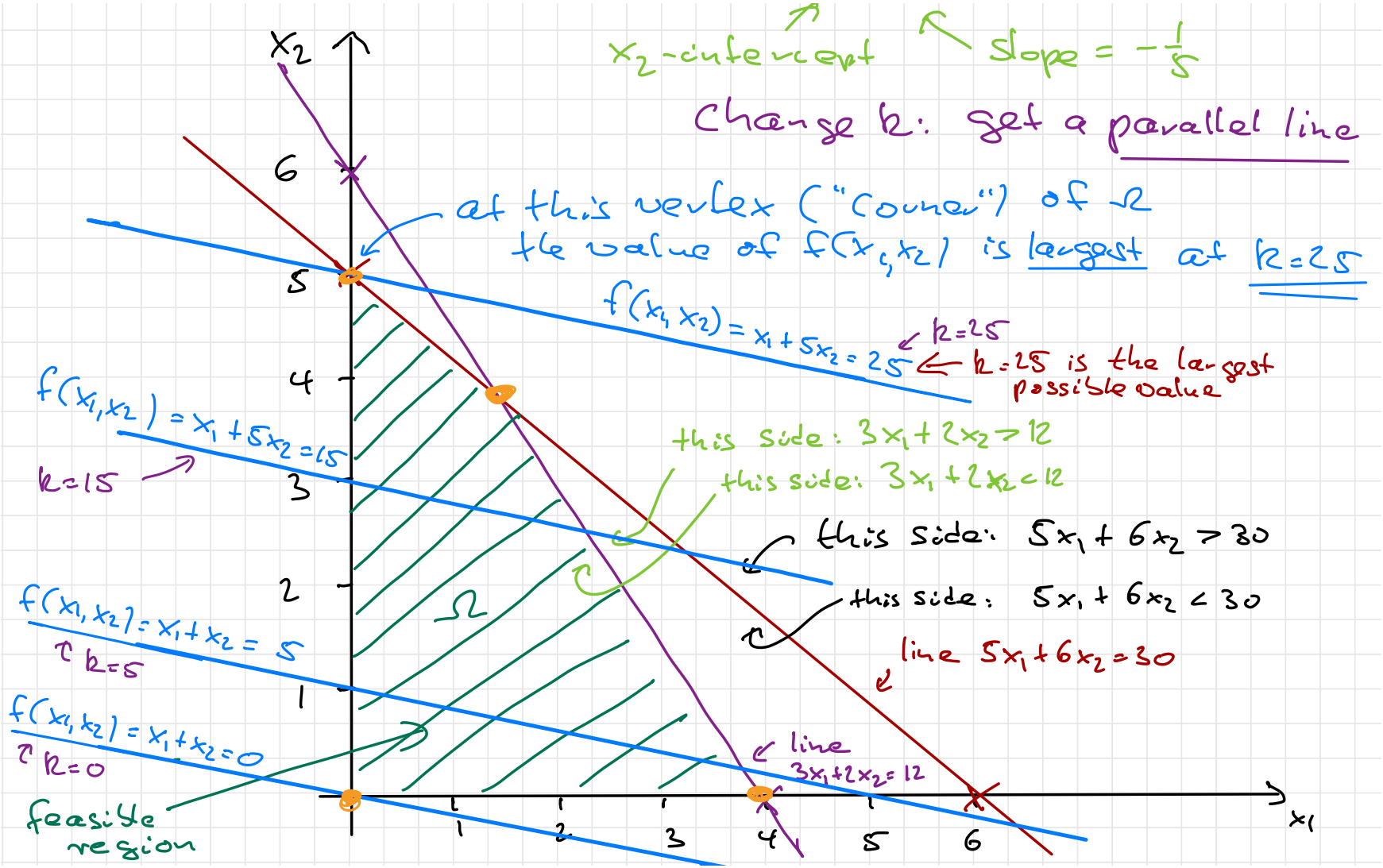
x_2 - intercept: $(x_1 = 0) \Rightarrow 2x_2 = 12 \Rightarrow \underline{\underline{x_2 = 6}}$

So we plot the x_1, x_2 intercepts of each line, and connect them.

$x_1 \geq 0, \; x_2 \geq 0 \Rightarrow$ <u>1st. quadrant</u>

objective function $f(x_1, x_2) = x_1 + 5x_2 = k = \text{const}$

$$\Rightarrow x_2 = \frac{k}{5} - \frac{x_1}{5}$$


x_2-intercept
slope $= -\frac{1}{5}$
Change k: get a parallel line
at this vertex ("corner") of Ω
the value of $f(x_1,x_2)$ is largest at $k=25$
$f(x_1,x_2) = x_1 + 5x_2 = 25$
$k=25$
$k=25$ is the largest possible value
$f(x_1,x_2) = x_1 + 5x_2 = 15$
$k=15$
this side: $3x_1 + 2x_2 > 12$
this side: $3x_1 + 2x_2 < 12$
this side: $5x_1 + 6x_2 > 30$
this side: $5x_1 + 6x_2 < 30$
line $5x_1 + 6x_2 = 30$
$f(x_1,x_2) = x_1 + x_2 = 5$
$k=5$
$f(x_1,x_2) = x_1 + x_2 = 0$
$k=0$
Ω
line $3x_1 + 2x_2 = 12$
feasible region
x_1
x_2
1
2
3
4
5
6

Let us sketch the feasible region Ω determined by the constraints:

The graphs of the equations

$$f(x_1, x_2) = x_1 + 5x_2 = k$$

(k =const) form parallel lines.

We want to find the line that intersects the feasible region, and has *largest* k.

From the graph we see that this line passes through the "corner point" $(x_1, x_2) = (0,5)$ of Ω, where the value of k is

$$f(0, 5) = 0 + 5 \cdot 5 = \underline{\underline{25}}$$

x_1 x_2

The answer is: $\mathbf{x} = \begin{bmatrix} 0 \\ 5 \end{bmatrix}$

The objective function $f(x_1, x_2) = x_1 + 5x_2$ has its max. value of 25 at the vertex (0,5) of the feasible domain